Bravais lattice：布拉维格子

The motif only contains one atom

Primitive cell：原胞

The smallest repeat unit that constructs lattice.

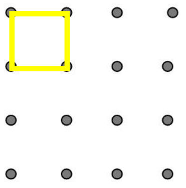
The way to choose primitive cell can be different, but **the volume of the primitive cell is constant**

Unit cell：晶胞

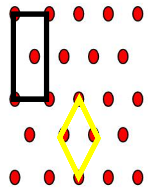
The smallest repeat unit **considering the symmetry of crystals**

Practice 1：choose 2D bravais lattice

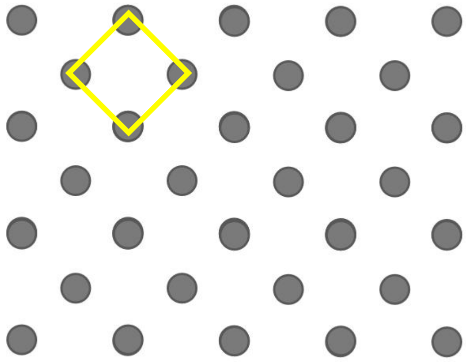
Plot the corresponding Bravais lattice/ unit cell of the following 2D patterns：



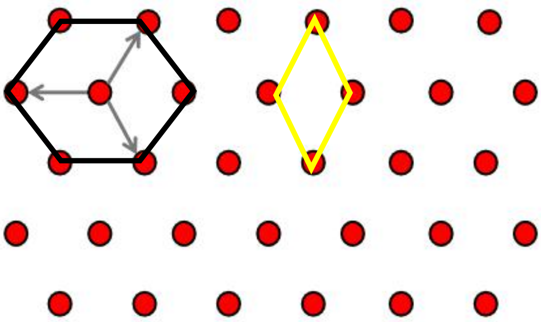
4mm($L^4$4P)



mm2($L^2$2P)

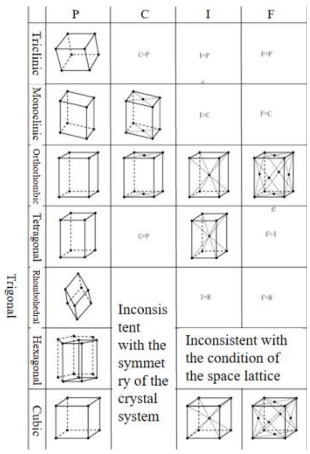


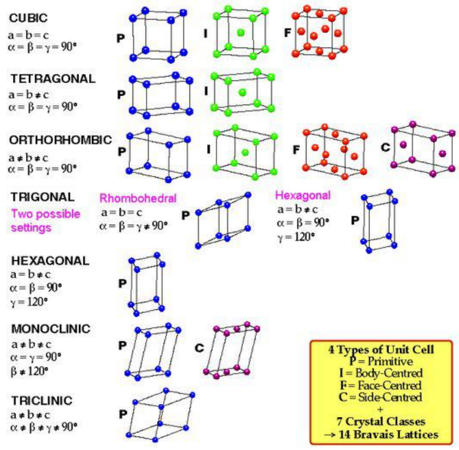
Practice 2：choose the unit cell of 2-D hexagonal lattice



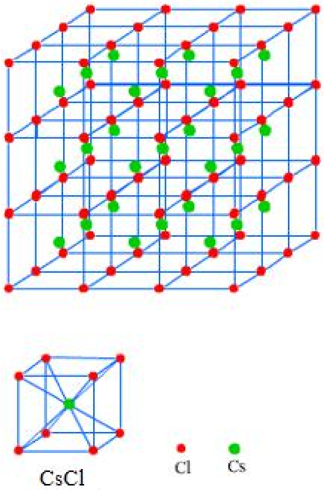
14 bravais lattices

Homework: Please derive 14 bravais lattices





14 bravais lattices



CsCl
Crystal Structure

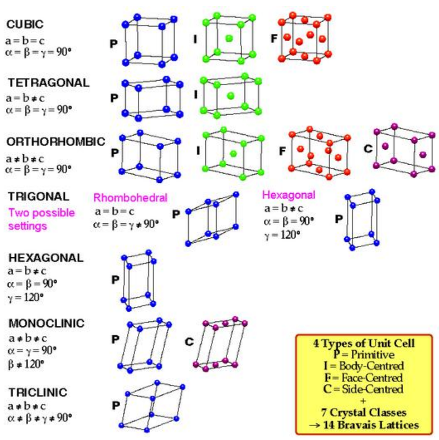
Practice 3: Please point out the lattice type of NaCl and CsCl crystals, respectively.

CsCl: Body Centered Cubic

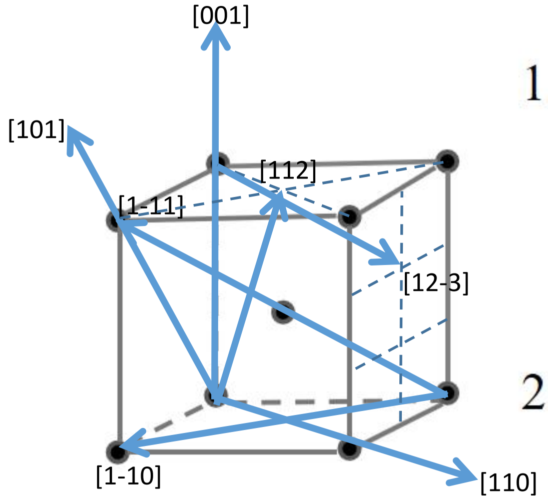
NaCl: Face Centered Cubic

HOMEWORK

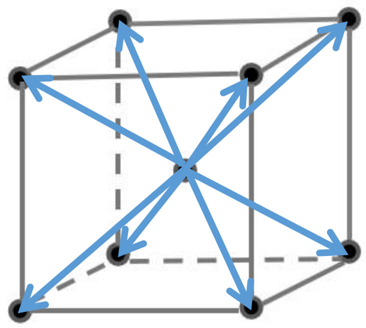
Please classify the 2-D Bravais lattices and point out the properties of their lattice parameters.



PRACTICE4: DENOTATION OF CRYSTALLOGRAPHIC DIRECTIONS



1. Please plot the directions in the left picture, [001],[101], [110], [1-10], [1-11], [112] and [12-3], respectively
2. Draw up all of the directions belong to the family of direction <111>.

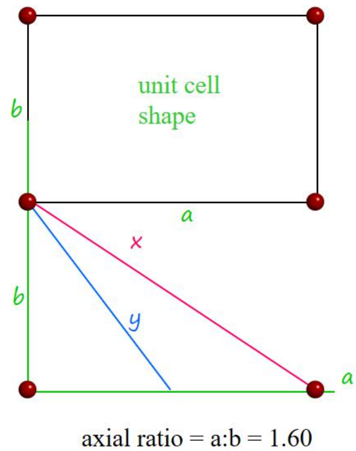


<111>

PRACTICE5: DENOTATION OF PLANE

x = (? ? ?)

y = (? ? ?)



(110)

(210)

PRACTICE5: DENOTATION OF PLANE

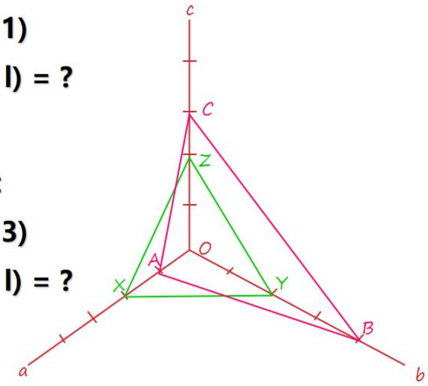
XYZ = (1 1 1)

ABC = (h k l) = ?

How about

XYZ = (1 2 3)

ABC = (h k l) = ?



(12 3 4)

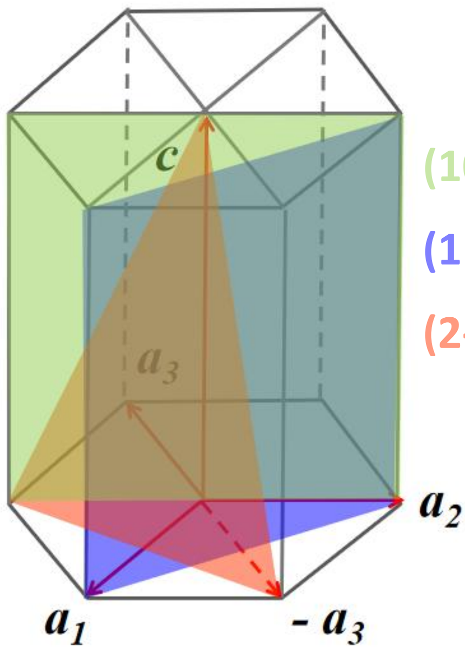
(2 1 2)

Index of directions and planes in hexagonal crystal system

Practice 8: denotation of planes

Q1. Please give the indices of the three colorful planes in the right picture based on three-axis and four-axis coordinate system, respectively;

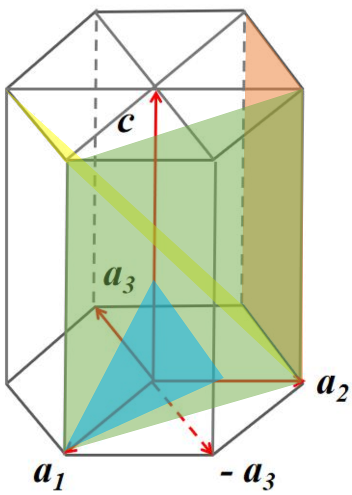
Q2. Please plot the planes, (-112), (-1100), (11-20), (123) and (12-33) in the hexagonal unit cell.



(100)(10-10)

(110)(11-20)

(2-11)(2-1-11)



(-112)

(-1100)=(-110)

(11-20)=(110)

(123)=(12-33)

Index of directions and planes in hexagonal crystal system

Green: [232][14-56]
Red: [041][-48-43]
Purple: [1-2-2][4-51-6]
Blue: [210][10-10]

Practice 9:

Q1. Please give the indices of the three colorful directions in the right picture based on the three-axis coordinate system and four-axis coordinate system, respectively.

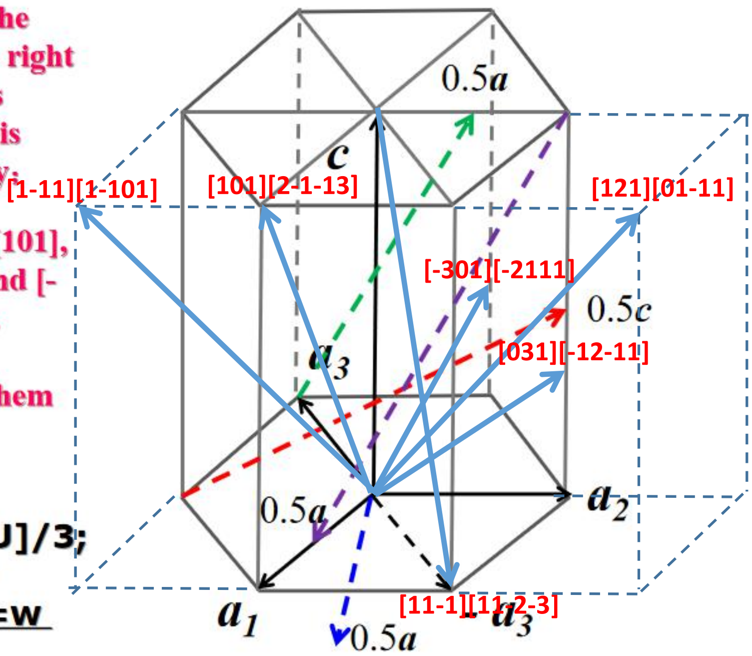
Q2. Please plot the directions [101], [11-1], [121], [1-11]，[-2111] and [-12-11] in the hexagonal lattice.

Q3. Point out in Q2 which of them belong to the same group of direction.

[11-1]&[1-11]
[-2111]&[-12-11]

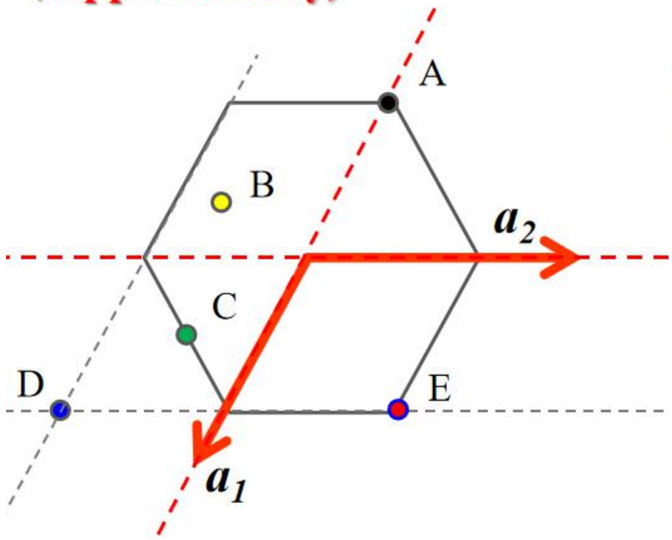
u=[2U−V]/3; v=[2V−U]/3; t=-[U+V]/3; w=W

U = u −t ; V= v −t ; W=w



Index of directions and planes in hexagonal crystal system

How to get the coordinates of a point in a Non - Cartesian coordinate system? (supplementory)



Practice:

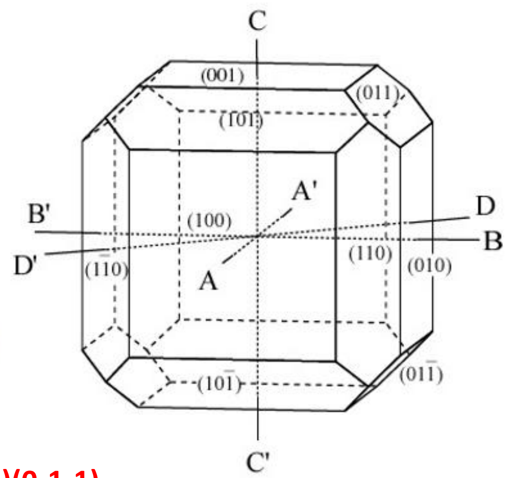
Give the coordinates of the points of A-E.

A: [[-10]]
B: [[-1/3-2/3]]
C: [[1/2-1/2]]
D: [[1-1]]
E: [[11]]

ZONE & ZONE AXIS

Exercise 10

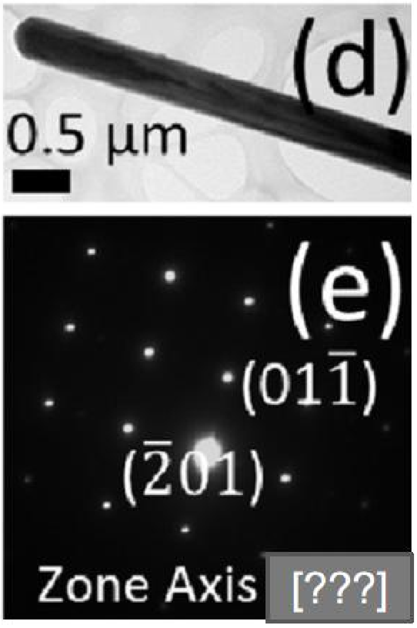
The right figure shows a cubic crystal with several naked facets.

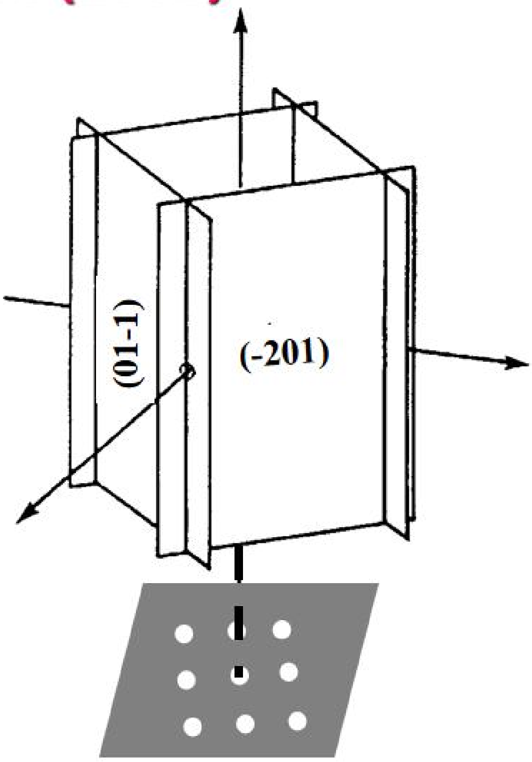


Please give the planes that belong to Zones AA', BB', CC' and DD'.

AA': (010)(0-10)(001)(00-1)(011)(01-1)(0-11)(0-1-1)
BB': (001)(00-1)(100)(-100)(101)(10-1)(-101)(-10-1)
CC': (010)(0-10)(100)(-100)(110)(-110)(1-10)(-1-10)
DD': (001)(00-1)(110)(-1-10)

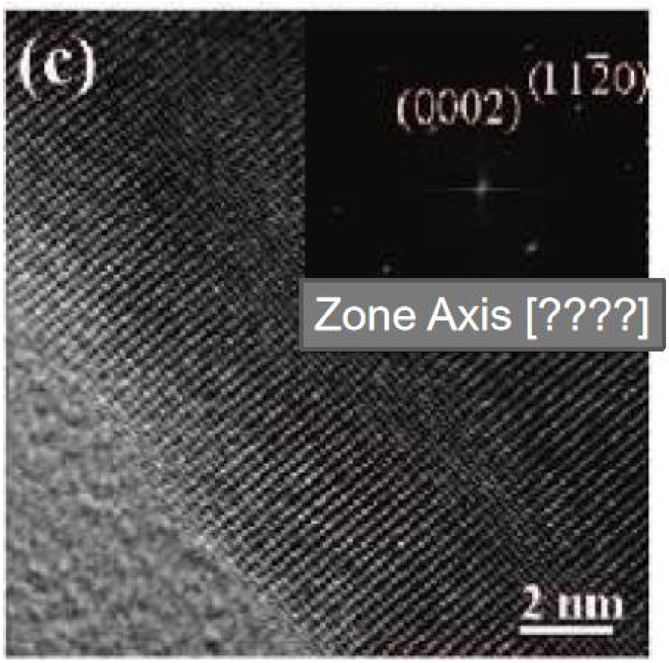
PRACTICE11: APPL. PHYS. LETT. 100, 103111 (2012)





[122]

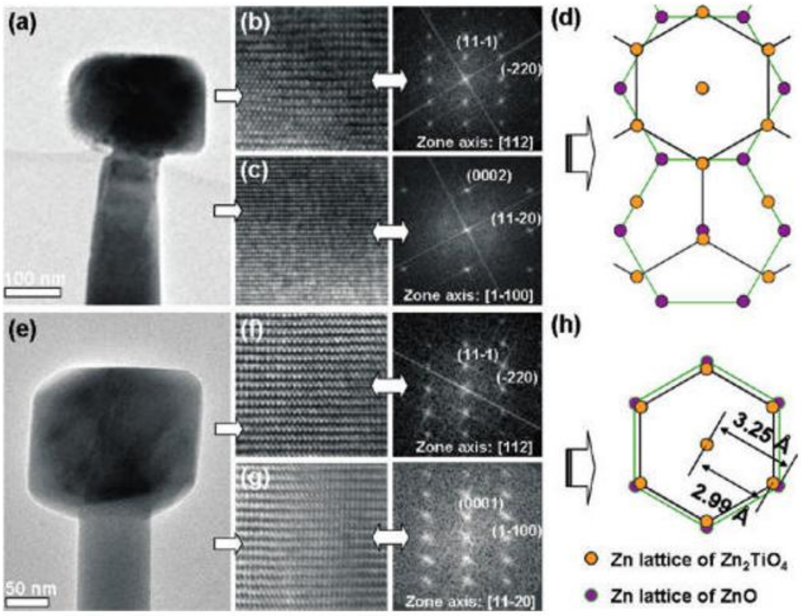
PRACTICE12: *ACS NANO 3 (2009) 53*



[1-10][1-100]

PRACTICE13: J. PHYS. CHEM. C 115 (2011) 78

Please give the calculation process to show how to get the zone axis in Figure b, c, f, and g.



$$\begin{vmatrix} u & v & w \\ 1 & 1 & -1 \\ -2 & 2 & 0 \end{vmatrix} = [112] \quad \begin{vmatrix} u & v & w \\ 0 & 0 & 2 \\ 1 & 1 & 0 \end{vmatrix} = [-110] = [1-100]$$

$$\begin{vmatrix} u & v & w \\ 1 & 1 & -1 \\ -2 & 2 & 0 \end{vmatrix} = [112] \quad \begin{vmatrix} u & v & w \\ 0 & 0 & 1 \\ 1 & -1 & 0 \end{vmatrix} = [110] = [11-20]$$